

AI & ML for Data Scientists

Class 5: Build a Model

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Why we need model?

1. Problems & Targets

2. Data Prep

3. Model Dev

4. Model Opt

5. Model Deploy

6. Improvement

Textbook & Coding

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Textbook & Coding

Why the companies will recruit you to build model?

- Prediction and Decision-Making for the Business
- Efficiency
- Precision
- → **Money** for You and Your Company

How to build a professional Model for business?

- For the data science, we usually have 6 steps to build a model:
 - Problem Definition and Data Collection
 - Data Preparation
 - Model Development
 - Model Evaluation and Optimization
 - Model Deployment
 - Continuous Improvement

Problem Definition and Data Collection

- **Understand the problem:** Define the objective and scope of the machine learning task.
- **Collect data:** Gather relevant data from various sources.

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Data Preparation

- **Preprocess data:** Clean and prepare the data for analysis.
- **Split the data:** Divide into training, validation, and test sets.
- **Normalization/Scaling:** Ensure data is on a consistent scale.
- **Feature engineering:** Create meaningful features from raw data.

Preprocess data

- Drop bad data
- Winsorize data
- Missing value imputation

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Model Development

- **Select an algorithm:** Choose a suitable machine learning algorithm.
- **Develop baseline models:** Create simple models for comparison.
- **Train the model:** Use the training data to fit the model.

Select an algorithm

- Supervised Machine Learning (OLS,LASSO,XGboost, ANN, CNN, RNN, Transformer)
- Unsupervised Machine Learning (KNN,PCA,Autoencoder)
- Reinforcement Learning (DQN,PPO)
- The model should fit your data and business need

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Model Evaluation and Optimization

- **Evaluate performance:** Test the model on the validation set.
- **Optimize hyperparameters:** Tune parameters to improve performance.
- **Address overfitting:** Use techniques like regularization or cross-validation.

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Model Deployment

- **Present your model:** Present your model to the business partners or seminars for academics like a salesman
- **Deploy to production:** Integrate the model into a real-world system.
- **Monitor performance:** Track the model's performance in production.
- **Handle feedback:** Use feedback to improve the model.

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Continuous Improvement

- **Periodic evaluation:** Regularly assess the model's accuracy.
- **Failure analysis:** Identify and address weaknesses.
- **Update and retrain:** Keep the model relevant with new data.

Conclusion

- Each stage is crucial for building effective models.
- DATA is huge important for the modeling
- Continuous improvement ensures long-term success.
- Don't forget to show your young talents to the audiences.

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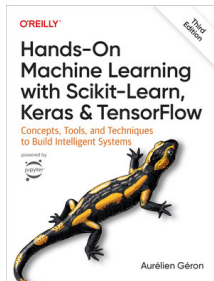
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Textbook & Coding

Textbook

- Hands-on Machine Learning with Scikit-Learn, Keras and TensorFlow
- Popular training textbooks for financial institutes, quants, or universities
- Pros: real practices
- Cons: lack math (The Elements of Statistical Learning for math)



Codes

- Codes: <https://github.com/ageron/handson-ml3/tree/main>
- Git Download: `git clone https://github.com/ageron/handson-ml3.git`

Read the textbook & run the codes

- Efficiently reading textbook and running codes are crucial for our modeling study
- Please follow me read the textbook and run the codes

Reference

1. Hands-on Machine Learning with Scikit-Learn, Keras and TensorFlow (3rd edition)
2. Andrew Ng's Coursera
3. Kaggle
4. Wikipedia
5. ChatGPT
6. DeepSeek